

Canonical identification of the one-dimensional jet coefficients

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Abstract

The jet coefficients $F_j(s) = e^{\beta s x_0} P_j(s)$ of the all-order 1D theorem were defined by finite polynomial algebra from the Taylor data of ξ, η . They are identified here with the normalised derivatives

$$F_j(s) = \frac{1}{j!} \partial_u^j (\eta(u) e^{\beta s \xi(u)}) \Big|_{u=0}, \quad j < R, \quad s \geq 0,$$

by uniqueness of Taylor coefficients: two polynomial approximations of the same function with $O(u^R)$ errors on $(0, b]$ agree below order R , and the amplitude jet remainder (unit 95) and Mathlib's Taylor theorem (unit 97) supply the two approximations. Consequently the jets do not depend on the truncation order. No chain rule or Faà di Bruno formula is used (Astra #4, rank 5). Zero `sorry/axiom`.

1 The things formalised

```
theorem coeff_eq_zero_of_isBig0 (b : ℝ) (hb : 0 < b) (R : ℕ) :
  (d : ℕ → ℝ) (K : ℝ), (u : ℝ) (Ioc (0 : ℝ) b, | n | ≤ n range R, d n * u ^ n | K * u ^ R) →
  n, n < R → d n = 0
theorem taylor_coeff_unique (b K K : ℝ) (hb : 0 < b) (R : ℕ) (f : ℝ → ℝ) (a a' : ℝ → ℝ)
  (h : u : ℝ) (Ioc (0 : ℝ) b, |f u - ∑ n range R, a n * u ^ n| ≤ K * u ^ R)
  (h : u : ℝ) (Ioc (0 : ℝ) b, |f u - ∑ n range R, a' n * u ^ n| ≤ K * u ^ R) :
  n, n < R → a n = a' n
theorem chartAmp_contDiff (s : ℝ) (R : ℕ) ( : ℝ → ℝ) (h : ContDiff ℝ R)
  (h : ContDiff ℝ R) : ContDiff ℝ R fun u => u * Real.exp ( * s * u)
theorem chartJet_eq_taylorData (b : ℝ) (h : 0 < ) (hb : 0 < b) (R : ℕ) (hR : 0 < R)
  ( : ℝ → ℝ) (h : ContDiff ℝ R) (h : ContDiff ℝ R) (s : ℝ) (hs : 0 ≤ s) :
  j, j < R →
  chartJet (taylorData ) (taylorData ) R j s
  = taylorData (fun u => u * Real.exp ( * s * u)) j
theorem chartJet_indep (b : ℝ) (h : 0 < ) (hb : 0 < b) (R R' : ℕ) (hR : 0 < R)
  (hRR' : ℝ → ℝ) ( : ℝ → ℝ) (h : ContDiff ℝ R') (h : ContDiff ℝ R')
  (s : ℝ) (hs : 0 ≤ s) :
  j, j < R →
  chartJet (taylorData ) (taylorData ) R j s
  = chartJet (taylorData ) (taylorData ) R' j s
```

2 Mathematics

If $\sum_{n < R} d_n u^n = O(u^R)$ on $(0, b]$, the constant term vanishes by letting $u \rightarrow 0^+$ (the polynomial is continuous with value d_0 at 0 and is squeezed to 0), and dividing by u the shifted sequence (d_{n+1}) is $O(u^{R-1})$; induction on R gives $d_n = 0$ for all $n < R$. Applied to the difference of two approximations

of the same f , this is uniqueness of Taylor coefficients. The chart amplitude $u \mapsto \eta(u)e^{\beta s \xi(u)}$ is C^R when ξ, η are, so unit 97's Taylor remainder bound gives one approximation with coefficients $\partial_u^j F(0, s)/j!$; unit 95's amplitude jet remainder gives the other, with coefficients $F_j(s) = e^{\beta s x_0} P_j(s)$ (constant $C(1+s)^R e^{\beta s(x_0 + L_1 b)}$, fixed $s \geq 0$).

3 Paper-facing meaning

The all-order 1D coefficients $c_j = k^{-1} \int s^{p+j/k-1} e^{-\beta s^2} F_j(s) ds$ now have their canonical, truncation-independent meaning: F_j is the j -th Taylor coefficient in u of the amplitude $\eta e^{\beta s \xi}$, exactly the object the paper's tree expansion enumerates. What remains for a literal match with `eq:flucttreeterms` is the finite combinatorial identity between $e_{j,m} = \sum_{\ell \leq j} y_\ell [u^{j-\ell}] (g_R^m)$ and the paper's order- j tree sum (Astra #4(f): recursion $\rightarrow$ partition formula $\rightarrow$ tree reindexing), none of which affects the asymptotic theorem.

4 Lean notes

- `Tendsto.mono_left ... nhdsWithin_le_nhds` needs the set named, `(nhdsWithin_le_nhds (s := Ioi 0))`, and the continuous function stated as a typed `have` (a bare `continuous_const.mul ...` leaves the constant a metavariable).
- `squeeze_zero_norm'` with `Ioc_mem_nhdsGT hb` handles the one-sided squeeze; `tendsto_nhds_unique` in `[>] 0` (NeBot instance found automatically) closes $d_0 = 0$.
- `Finset.sum_range_succ'` peels the constant term; `le_of_mul_le_mul_left` divides the $O(u^{R+1})$ bound by u .
- $\leftarrow$ `Finset.sum_sub_distrib` needs the subtraction of sums adjacent; reorder with a `show ...` by `ring` first.